DHAYALAN K

MERN Stack Developer | MongoDB, Express.js, React, Node.js and Cloud Deployments

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IMPACT AT A GLANCE

MERN Architecture Specialist React v18, Express REST APIs, Node.js Async Runtimes & MongoDB Atlas | **92% Merit Score** in Certified MongoDB Data Modeling & Schema Design | **AWS Container Certified** Amazon ECS Knowledge Badge & Cloud Deployments | **Mobile-First Web Products** shipped with zero-jank LocalStorage cart state & sub-second load times

Specialized MERN Stack Developer with end-to-end expertise in constructing scalable single-page applications and high-throughput REST APIs. Skilled in designing decoupled React architectures with modern hooks and state management, building asynchronous Express/Node.js backends with robust middleware security, and modeling performant NoSQL data structures in MongoDB (certified with 92% merit). Experienced in building consumer web applications, integrating client storage mechanics, and running containerized Node applications with Docker.

MERN STACK ARCHITECTURE & ENGINEERING

React Frontend & Component Engineering

- Engineered single-page application interfaces utilizing React functional components, custom hooks (`useCallback`, `useMemo`), and Context API for global state synchronization.
- Built **Shahihennaholic Storefront & Cart Engine**: client-side e-commerce platform with persistent LocalStorage synchronizer ensuring cart items persist across page navigation.
- Implemented responsive styling systems with Tailwind CSS, achieving 100% mobile-first compatibility, zero layout shift, and instant asset rendering.
- Engineered **FLARUM1**: independent modular JavaScript companion extension for single-page forum interfaces, hooking into routing events and component mounts.

Express & Node.js API Layer & MongoDB Modeling

- Structured modular Express.js backends with decoupled routing controllers, custom error handlers, and input validation pipelines.
- Configured MongoDB Atlas collections, Mongoose ODM schemas, compound indexes, and aggregation pipelines (awarded 92% Merit Score).
- Containerized Node.js services and MongoDB replica sets using Docker and Docker Compose for reproducible cloud environments.

TECHNICAL SKILLS

Frontend (React Ecosystem): React (v18+), Functional Components, Custom Hooks, Context API, Next.js (App Router), Tailwind CSS, JavaScript (ES6+), HTML5 Semantic UI, CSS3 Flexbox/Grid

Backend (Node.js & Express): Node.js Runtime, Express.js Framework, RESTful Routing, Middleware Pipeline, JWT Authentication, CORS, Asynchronous Programming (Async/Await), JSON API

Database (MongoDB): MongoDB Atlas, Mongoose ODM, Schema Validation, Aggregation Pipelines, Compound Indexing, Performance Tuning (92% Score)

DevOps & Containerization: Docker, Docker Compose, Amazon ECS (Fargate), Linux (Ubuntu), Nginx Reverse Proxy, Git, GitHub Actions, NPM/Yarn

API Testing & Tooling: Postman (Collection Testing, Environment Variables), Vite, Webpack, Chrome DevTools Profiling

PROFESSIONAL EXPERIENCE

Full-Stack MERN Developer, Web Platforms & Client Projects **2024 to Present, Coimbatore**

- Engineered responsive, database-backed web applications using React, Node.js, Express, MongoDB, and modern utility-first CSS systems.
- Developed RESTful endpoints with input validation, JWT token authorization, and structured MongoDB queries.
- Integrated client-side state persistence workflows with LocalStorage, supporting uninterrupted customer sessions.

Systems & Automation Engineer, Independent Systems Lab **2025 to Present, Coimbatore**

- Containerized Node.js microservices and local MongoDB instances using Docker, ensuring environment parity across machines.
- Constructed Python event buffer daemons to batch client interactions prior to document store serialization.

KEY PROJECTS & CODEBASES

Shahihennaholic Storefront & Cart Engine github.com/DHAYALANKANIAPPAN/PROJECTS

- Architected client-side e-commerce catalog with persistent LocalStorage synchronizer and mobile-optimized ordering.
- Integrated automated checkout flow generating pre-formatted WhatsApp message payloads. Stack: JavaScript, HTML5, CSS3, LocalStorage.

FLARUM1 — Modular Community UI Extension github.com/DHAYALANKANIAPPAN/flarum

- Engineered independent JavaScript extension package hooking into single-page application router events and component mounts.
- Injected custom responsive themes and interactive controls without modifying upstream source code. Stack: JavaScript, Mithril.js.

Browser Telemetry & Event Interceptor Pipeline github.com/DHAYALANKANIAPPAN/PROJECTS

- Built real-time event stream tool in Python capturing DOM mutations, event queues, and telemetry streams formatted for JSON document ingestion.

EDUCATION

Bachelor of Engineering in Computer Science and Engineering (Cyber Security), Pursuing

Sri Shakthi Institute of Engineering and Technology, Coimbatore (2024 – 2028) | Grade Point Average: 8.21 out of 10.0.
Available for full-time employment & internships.

VERIFIED CERTIFICATIONS, AWARDS & ACTIVITIES

- Certificate of Merit: Java + Mongo DB (92% Score) — Hazhtech Career Development Centre / IACT (NoSQL schema modeling).
- Amazon ECS Knowledge Badge Assessment — Amazon Web Services (AWS) · Credly Official.
- Deploying a CI/CD Pipeline using AWS CDK — AWS Skill Builder Official Lab.
- Yukthi CTF 2.0 Participant — National-Level Capture The Flag competition, Tamil Nadu Police.
- 0xTi - HackTiVate 24-Hour CTF Participant — Rajalakshmi Engineering College, Chennai.